

Valentin Nigolian, PhD

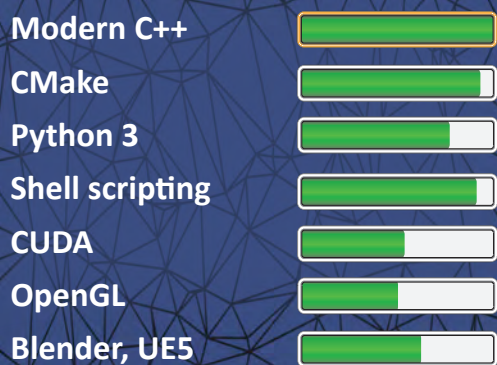


<https://vnigolian.github.io/>
valentin.nigolian@gmail.com
+41 76 360 62 55
Biel/Bienne, Switzerland

EDUCATION

- 2019 **M.Sc. Computer Science**
Ecole Polytechnique Fédérale de Lausanne
- 2016 **B.Sc. Computer Science**
Ecole Polytechnique Fédérale de Lausanne
- 2010 **Swiss Matura**
Geneva

TECHNICAL SKILLS



SOFT SKILLS



LANGUAGES



PROFILE

Computer scientist specializing in geometry processing and numerical optimization with a track record of innovative algorithm design, and published in top-tier venues.

EXPERIENCE

- 2024 - 2026 **Postdoctoral Researcher** Computer Graphics Group
University of Bern
Switzerland
Innovative simplicial map generation through non-conforming meshing
- 2020 - 2024 **Ph.D. - Digital Geometry** Computer Graphics Group
University of Bern
Switzerland
"Robust Tetrahedral Maps - The Shrink-and-Expand Framework"
Supervised by Prof. D. Bommes
Summa Cum Laude
- 2020 - 2025 **Teaching Assistant** Computer Graphics Group
University of Bern
Switzerland
Digital Geometry,
Applied Optimization
- 2019- 2020 **Software Engineer** GaitUp
Lausanne, Switzerland
Led the architecture and development of a cross-platform gait analysis SDK in C++, deployed across desktop, mobile, and wearable devices for athletic and medical applications.

PUBLICATIONS

- 2026 **SIGGRAPH, SIGGRAPH ASIA, SIGGRAPH ASIA**
2024
2023 "Robust Deformation-based 3D Bijective Mapping through Differential Boundary Shape Matching"
"Expansion Cones: A Progressive Volumetric Mapping Framework"
"A Progressive Embedding Approach to Bijective Tetrahedral Maps driven by Cluster Mesh Topology"
Complex, multi-library algorithmic development in C++.
- 2019 **UIST**
"INVANER: Interactive VAscular Network Editing and Repair."
Unreal-Engine-powered interactive interface prototype developed with and for medical doctors in Tokyo.
- 2017 **ROMAN**
"Self-Reconfigurable Modular Robot Rnterface Using Virtual Reality."
VR-augmented, C++-based interface for forward-kinematics robots pose generation.